CS 630: Software Testing and Quality Assurance

3-2 Activity: Breaking Down Features for Test and Planning

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Test Plan Excerpt: Login Feature for a Mobile Application

For this activity, I selected my TraffickAid mobile application because it includes several important features that require thorough testing to ensure reliability, security, and usability. Since the application allows users to report incidents, access resources, communicate within the community, and manage their accounts, it is essential that each feature functions correctly under both normal and unexpected conditions. For this test plan excerpt, I focused on the login feature, as it is the gateway to secure access and plays a critical role in protecting user information and providing a positive user experience.

Decompose the Feature into Multiple Testable Units

Unit of Measurement

The primary unit of measurement selected is test cases.

Reason for Selection

Test cases provide a structured way to verify that each part of the login process functions correctly. They allow testers to validate both expected behavior and error conditions while making it easy to measure progress, coverage, and software quality.

Testable Unit

Unit	Expected Function
Login Screen	Displays username/email and password fields with Login button.
Username/Email Validation	Accepts valid usernames or email addresses and rejects invalid formats.
Password Validation	Accepts valid passwords while enforcing password rules.
Authentication Service	Verifies user credentials against the database.
Forgot Password	Sends password reset instructions to registered users.
Session Management	Creates a secure user session after successful authentication.
Logout Function	Ends the active session and redirects the user to the login page.
Error Handling	Displays clear error messages when login fails
Security Controls	Prevents unauthorized access through account lockout and secure authentication mechanisms.

Define Measurable Metrics for the Test Effort

Dashboard-Level Metrics

Metric	Target
Test Case Coverage	100% of planned login test cases executed
Requirements Coverage	100%
Defect Density	Fewer than 2 critical defects
Open Critical Bugs	0 before release

Go/No-Go by Unit

Unit	Criteria
Login Screen	All interface elements are displayed correctly
Authentication	Users successfully authenticate with valid credentials
Password Reset Session Management	Reset emails are successfully delivered Session is created and terminated correctly
Security Controls	No critical security vulnerabilities remain

A No-Go decision will occur if any critical functionality fails, or security issues remain unresolved.

Pass/Fail Percentage

- Minimum passing rate: 95%
- Critical authentication test cases: 100% must pass
- High-priority security test cases: 100% must pass

Performance Thresholds

- Login response time should be less than 2 seconds under normal conditions.
- Password reset requests should be completed within 5 seconds.
- The system should support at least 500 concurrent users without noticeable performance degradation.
- CPU and memory utilization should remain within acceptable limits during testing.

Usability Checks

- Login page is easy to navigate.
- Buttons are clearly labeled.
- Error messages are understandable.
- Password field masks entered characters.
- The keyboard automatically appears when input fields are selected.
- Application is accessible using screen readers and proper color contrast.

Identify Edge Cases Worth Testing

The following edge cases should be included during testing:

Edge Case	Expected Result
Empty username and password	User receives validation message
Empty password only	Login is prevented.
Empty username only	Login is prevented.
Invalid email format	Error message is displayed
Incorrect password	Authentication fails
SQL injection attempt	Input is rejected, and application remains secure.
Extremely long username (500+ characters)	System handles input safely without crashing
Password containing special characters	Accepted if it meets security requirements.
Network disconnect during login	User receives an appropriate connection error.

Multiple failed login attempts	Account lockout activates after configured threshold
Session timeout	User is redirected back to the login page.
Login after password reset	User successfully authenticates using the new password.
Simultaneous login from multiple devices	Application follows the expected session management policy

Structured Test Plan Excerpt

Test Unit	Test Metric	Edge Cases
Login Screen	100% UI coverage	Empty fields
Email Validation	100% valid/invalid input testing	Invalid formats, long inputs
Password Validation	100% password rule coverage	Special characters, empty password
Authentication	100% credential verification	Incorrect credentials, SQL injection
Forgot Password	Email delivery success rate	Invalid email, expired reset link
Session Management	Session creation and expiration	Timeout, multiple devices
Logout	Successful session termination	Browser refresh after logout
Security Controls	Zero critical vulnerabilities	Brute-force attacks, account lockout

How Edge Cases Improve Test Coverage

Testing edge cases significantly improves software quality because it verifies how the application behaves under unusual or unexpected conditions. While standard test cases confirm that the login feature works correctly during normal use, edge cases help uncover defects that may otherwise remain hidden until the application is deployed. For example, testing empty fields, invalid input formats, SQL injection attempts, network interruptions, and multiple failed logins attempts to help validate the application's input validation, security, and error handling. These scenarios reduce the likelihood of authentication failures, unauthorized access, application crashes, and poor user experiences. Including edge cases creates a more comprehensive test plan by ensuring that both common and uncommon user behaviors are evaluated before release. As a result, the software becomes more reliable, secure, and better prepared for real-world usage.